



GRANDICE DOMAIN INTELLIGENCE

TECHNICAL WHITE PAPER • VERSION 1.0

Financial Analysis Agents

Deterministic, source-attributed decision support with bounded authority and review-gated outputs.

An implementation-oriented architecture for portfolio concentration, bounded risk assessment, and historical company financial analysis.

CALCULATE

DISCLOSE

REVIEW

PUBLISHED

September 2026

METHODOLOGY

grandice-domain-1.0.0

STATUS

Initial implementation

Controlled financial analysis—not autonomous financial advice.

Grandice provides three constrained financial-analysis agents for portfolio concentration, bounded risk assessment, and historical company-ratio analysis. Each agent dispatches a strict, typed request to predefined deterministic calculations.

Inputs include caller-provided source metadata and effective dates. Outputs include methodology version, source citations, assumptions, limitations, warnings, and an invariant requirement for professional review. The architecture deliberately excludes open-ended objectives, web browsing, live market-data retrieval, order execution, account modification, and model-generated tool arguments.

Grandice Privacy Shield can protect data locally in the client SDK or sidecar before transfer, and the gateway applies a second residual policy scan before financial schema validation. Detected categories are rejected before calculation, while domain audit records retain operational metadata without request or response bodies.

CORE THESIS

The useful role of an agent in high-stakes financial workflows is not unlimited autonomy. It is controlled workflow selection, reproducible calculation, explicit provenance, and an enforceable handoff to qualified human judgment.

Standardize repeatable analysis without hiding the decision boundary.

Financial teams frequently inherit calculations spread across spreadsheets, scripts, and analyst-specific conventions. Source dates become detached from outputs, formula changes are difficult to trace, and generative systems can add nondeterministic interpretation where repeatability matters.

Formula drift

Thresholds and denominator rules diverge between teams and versions.

02

Detached provenance

Results circulate without the source identifier or effective date that supported them.

03

Excessive authority

Broad agent permissions conflict with least-authority controls in regulated workflows.

04

Content-heavy audit

Logging full requests can create a second repository of confidential financial data.

Grandice addresses the calculation and control-plane portion of this problem. It converts caller-supplied, dated facts into reproducible analytical outputs under declared schemas and methodology versions. The consuming institution remains responsible for source quality, validation, supervision, suitability, communications, and any resulting action.

02

SCOPE AND RESPONSIBILITY BOUNDARY

Layered privacy and bounded financial authority.

“Financial Advisor Agent” is used as an umbrella product term for a constrained analysis suite and developer control plane. Financial judgment, suitability, compliance approval, and any resulting action remain with qualified professionals and the adopting institution’s governed processes.

Implemented

- Exact-decimal portfolio arithmetic
- Transparent bounded risk scoring
- Historical company-ratio calculations
- Caller-attributed source metadata
- Local SDK or sidecar protection before transfer
- Gateway residual scanning before financial validation
- Body-free domain audit metadata
- Mandatory professional-review flag

Outside current authority

- Personalized buy, sell, hold, product, or allocation advice
- Suitability, fiduciary, or regulatory-compliance determinations
- Live market data, browsing, forecasts, or source verification
- Order routing, trading, rebalancing, or account modification
- Model-generated tool calls or unrestricted agent objectives
- Automatic approval or downstream action without professional review

03

DESIGN PRINCIPLES

Six controls shape the system.

01

Deterministic financial arithmetic

Identical valid inputs and methodology versions produce identical calculation outputs.

02

Bounded agent authority

Each registered agent accepts one declared schema and invokes only predefined calculations.

03

Source-attributed inputs

Fact-bearing records reference caller-provided source identifiers and explicit as-of dates.

04

Data minimization

Identity fields, unrestricted objectives, and free-form personal profiles are excluded from financial schemas.

05

Metadata-oriented auditability

Receipts retain execution facts while excluding financial request and response bodies.

06

Mandatory professional review

Every analysis response carries `professional_review_required: true`.

A fail-closed request path for configured controls.

The gateway authenticates and bounds the request before financial validation. Raw JSON is inspected for configured sensitive-data patterns, then validated against a strict domain schema. Only validated inputs reach the deterministic calculation layer.

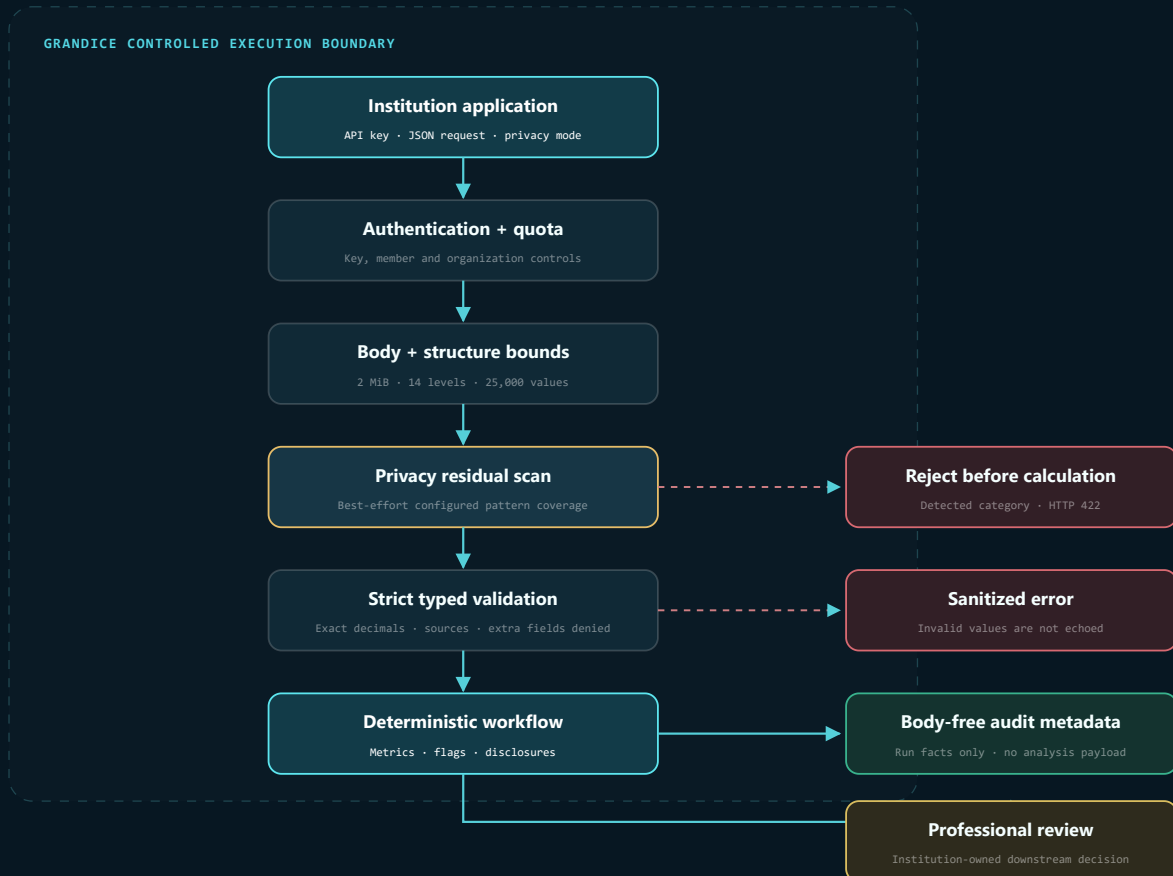


Figure 1. Controlled analysis lifecycle. No browsing, live feed, model-generated tool arguments, or trade execution occurs within the boundary.

One declared schema maps to one deterministic workflow.

In this release, “agent” means a registered typed workflow—not an open-ended planner. Tool names in the registry describe fixed calculation capabilities. They do not represent model-selectable external actions.

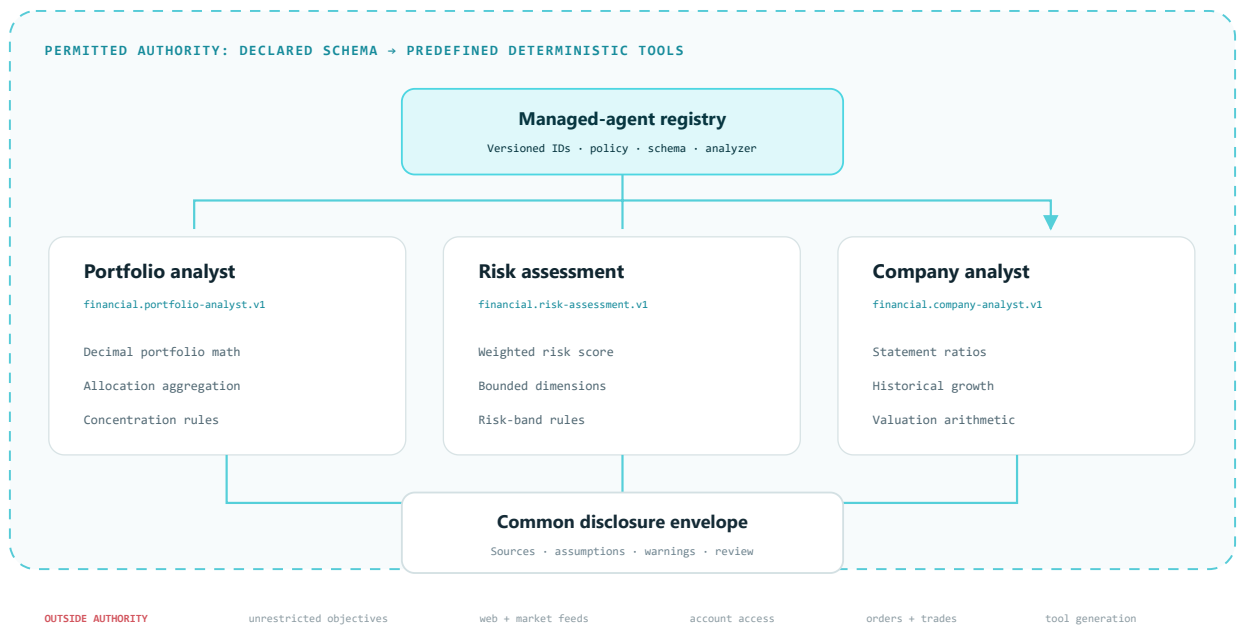


Figure 2. Agent authority is bounded by code and schema. The common envelope does not convert screening results into approved recommendations.

PORTFOLIO

Portfolio Analysis Agent

financial.portfolio-analyst.v1

Calculates weights, allocation by asset class, sector and country, top concentration, HHI, effective positions, and transparent diversification flags.

Excludes correlation, volatility, liquidity, tax, fees, transaction costs, look-through exposure, and allocation recommendations.

RISK

Risk Assessment Agent

financial.risk-assessment.v1

Applies disclosed fixed weights to five caller-scored dimensions bounded from one through five, then maps the score to a transparent band.

Not a psychometric instrument, suitability determination, or product-selection mechanism.

COMPANY

Company Financial Analysis Agent

financial.company-analyst.v1

Calculates historical growth, margins, net debt, leverage, and supported valuation multiples from two or more cited financial periods.

Excludes forecasts, peer normalization, DCF, accounting adjustments, and valuation opinions.

Inspectable formulas replace opaque model arithmetic.

6.1 Portfolio concentration and allocation

For each position i , market value V_i is divided by the total portfolio market value. Category allocations aggregate values before division.

TOTAL MARKET VALUE

$$V_{\text{total}} = \sum V_i$$

POSITION WEIGHT

$$w_i = V_i / V_{\text{total}}$$

CATEGORY WEIGHT

$$w_c = \sum_{i \in c} V_i / V_{\text{total}}$$

HERFINDAHL-HIRSCHMAN INDEX

$$HHI = \sum w_i^2$$

EFFECTIVE POSITIONS

$$N_{\text{effective}} = 1 / HHI$$

Effective positions express concentration equivalence, not economic diversification. Ten holdings do not imply ten effective positions when value is concentrated in a smaller subset.

< 5 POSITIONS	> 25% SINGLE POSITION	> 0.18 HHI	> 80% SINGLE ASSET CLASS
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Current screening thresholds use unrounded values. Rounded output is presentation only. Triggered rules are indicators—not instructions to rebalance.

6.2 Bounded risk scoring

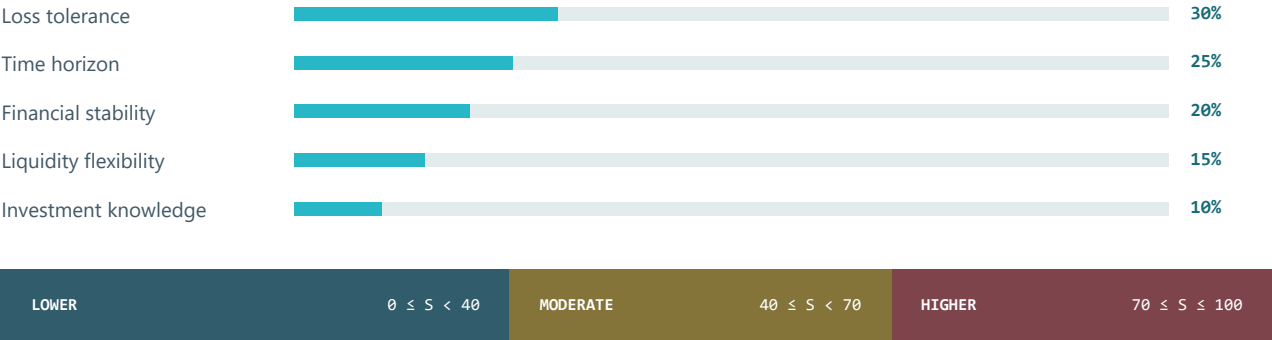
Each caller-scored rating r_j is an integer from one to five. Ratings are normalized to a zero-to-one-hundred scale and combined with fixed disclosed weights.

NORMALIZED COMPONENT

$$n_j = ((r_j - 1) / 4) \times 100$$

AGGREGATE SCORE

$$S = \sum \alpha_j n_j$$



Weights and bands are transparent policy choices. Transparency alone does not establish empirical validity or suitability for a population.

6.3 Company financial analysis

Historical growth

$$(\text{Current} - \text{Prior}) / |\text{Prior}|$$

Omitted when the prior value is zero

Margin

$$\text{Profit measure} / \text{Revenue}$$

Gross, operating, net, and EBITDA where supplied

Net debt

$$\text{Total debt} - \text{Cash}$$

Net-debt-to-EBITDA only when EBITDA is positive

Market capitalization

$$\text{Diluted shares} \times \text{Share price}$$

Requires both latest-period inputs

Enterprise value

$$\text{Market cap} + \text{Total debt} - \text{Cash}$$

Point-in-time arithmetic

Valuation multiples

$$\text{P/S} \cdot \text{P/E} \cdot \text{EV/EBITDA}$$

Reported only when denominators support them

Source attribution is validated; source truth is not inferred.

Each analysis carries an analysis date and one or more source records. Fact-bearing positions, risk profiles, and financial periods reference a declared opaque source identifier.

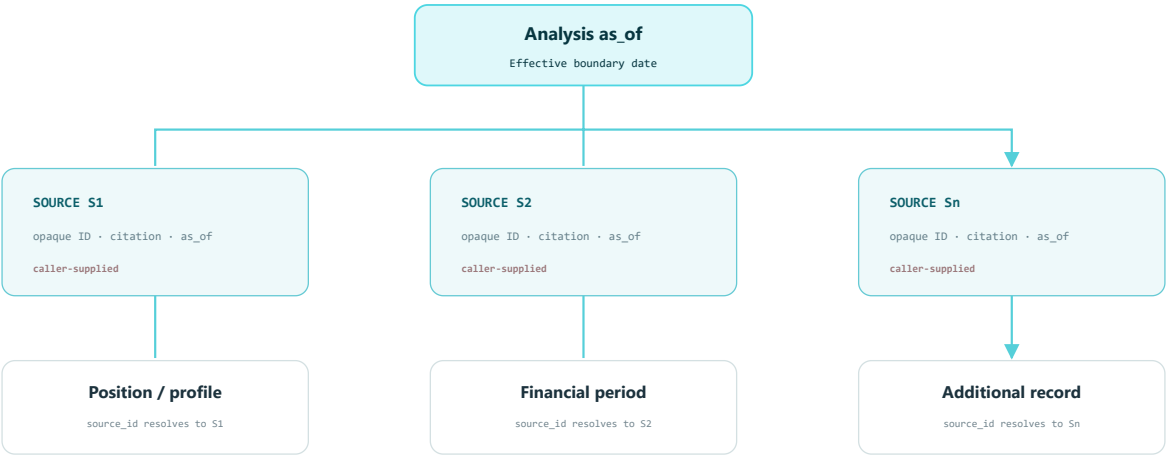


Figure 3. Unique and resolvable references plus temporal consistency are enforced. Grandice does not authenticate a publication, verify its contents, or determine economic comparability.

Unique source IDs	Resolvable references	Source date ≤ analysis date	Period date ≤ source date
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Layer controls, then describe their limits precisely.

The strongest workflow protects sensitive values locally in the client SDK or loopback sidecar. The domain gateway still performs a residual pattern scan over raw JSON before schema validation. Detected categories cause rejection rather than mutation of typed financial inputs.

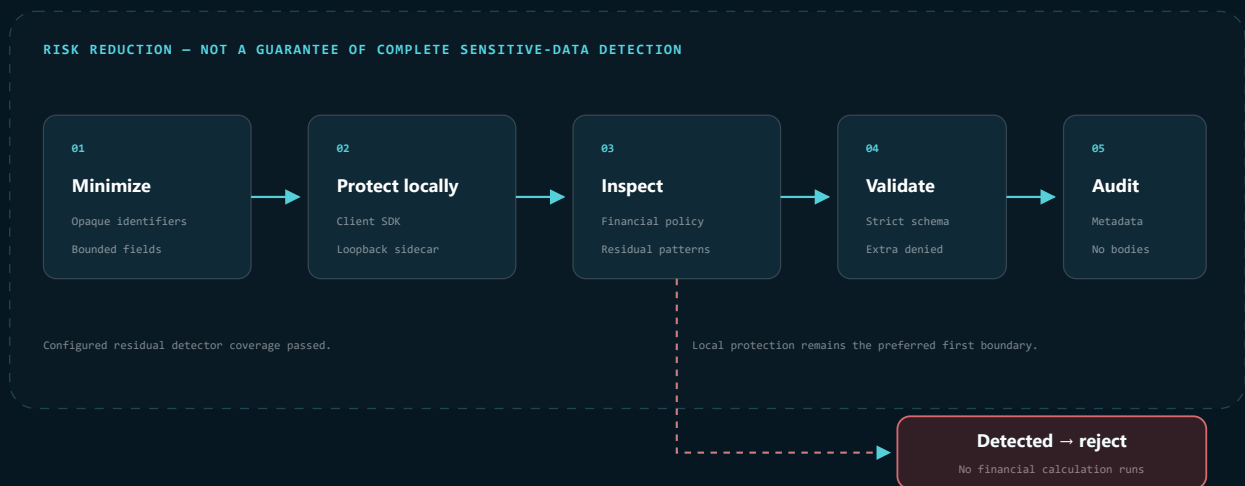


Figure 4. Defense in depth combines local protection, residual inspection, schema minimization, and metadata-only domain audit.

PREFERRED

Client or sidecar tokenization before transfer

SERVER CONTROL

Best-effort configured detector coverage

FAILURE MODE

Reject detected residual values before calculation

AUDIT SCOPE

Domain table excludes request and response bodies

Record execution facts without making the audit table a data lake.

Each completed or rejected domain workflow writes a metadata-oriented audit record. Retrieval is scoped to the API key that created the run. The domain record does not contain holdings, questionnaire ratings, citations, monetary values, or analysis output.

DOMAIN RUN RECEIPT

dom_.....

TIMESTAMP UTC execution time	ENDPOINT Financial route
AGENT Versioned agent ID	PRIVACY MODE off / client / sidecar / hosted
STATUS + DURATION Outcome metadata	SCHEMA NAMES Input + output contract
COUNTS Sources + warnings	METHODOLOGY grandice-domain-1.0.0

EXCLUDED

request body · response body · citations · financial values · generated analysis

“Body-free domain audit” describes the Grandice domain_runs record. It does not characterize every surrounding reverse proxy, application, infrastructure, security, or client log.

Calculation ends where accountable judgment begins.

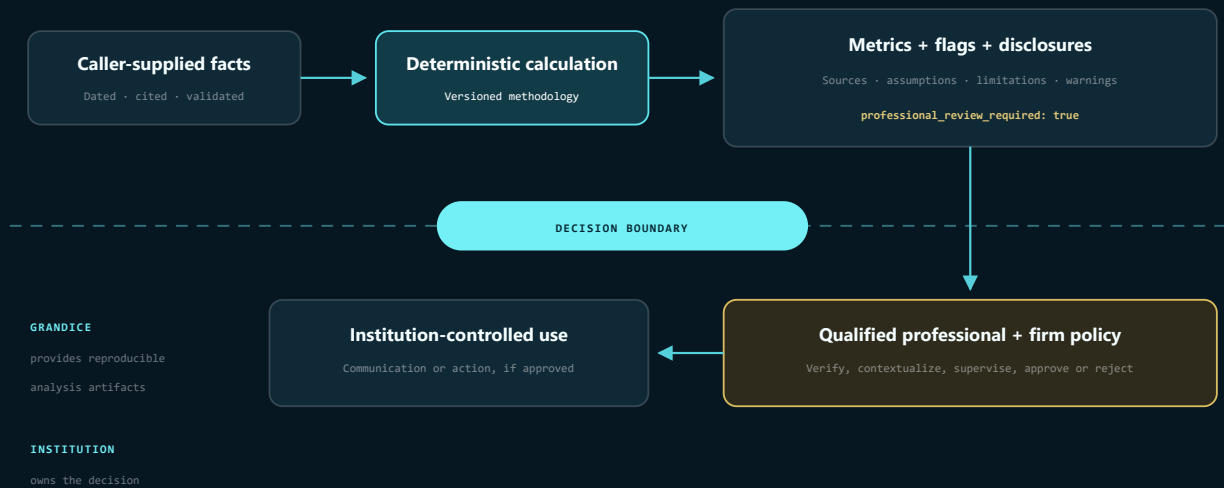


Figure 5. The professional-review flag is a response invariant. The current API does not perform, verify, or record completion of institutional review.

Recommended downstream review

1. Confirm source identity, dates, completeness, and authority.
2. Reconcile totals and period values to systems of record.
3. Validate classifications, currency consistency, and denominator rules.
4. Review triggered flags, omitted ratios, assumptions, and limitations.
5. Apply firm-approved suitability, supervisory, legal, and compliance processes.
6. Record reviewer identity, disposition, and any decision in the institution's system.

Embed analysis inside existing controlled workflows.

ADVISER WORKSTATION Concentration screening A portfolio system submits dated, locally tokenized holdings using opaque instrument IDs. Grandice returns allocation and concentration indicators. The adviser determines whether further analysis or communication is appropriate.
RISK INTAKE Transparent questionnaire scoring A firm-approved questionnaire maps responses to five bounded ratings. Grandice calculates the disclosed score and band. A separate supervisory workflow decides how, or whether, the result may be used.
EQUITY RESEARCH Historical ratio pipeline An analyst or controlled ETL process supplies at least two cited periods. Grandice computes supported ratios. Analysts reconcile values and add qualitative, forecast, peer, and market context outside the service.
DEVELOPER PLATFORM Direct API or registry dispatch Applications can call a known financial endpoint or discover and invoke a managed agent through a common envelope. Both paths preserve request ID, methodology, sources, warnings, and review requirements.

DIRECT	POST /v1/financial/portfolio/analyze
DIRECT	POST /v1/financial/risk-assessment
DIRECT	POST /v1/financial/company/analyze
REGISTRY	GET /v1/agents
DISPATCH	POST /v1/agents/runs
AUDIT	GET /v1/domain/runs/{request_id}

Resource and schema limits narrow the operating envelope.

2 MiB MAXIMUM DOMAIN REQUEST	14 MAXIMUM JSON LEVELS
25,000 MAXIMUM JSON VALUES	5,000 MAXIMUM POSITIONS
100 MAXIMUM PERIODS OR SOURCES	34 MAXIMUM DECIMAL DIGITS

Exact decimal contract Monetary values arrive as JSON strings or integers. Binary JSON floats, non-finite numbers, excessive magnitude, and excessive scale are rejected.
Strict schema contract Extra fields, duplicate source IDs, unresolved references, inconsistent dates, duplicate instruments, and duplicate periods are rejected.
Sanitized failures Validation responses describe field locations and rules without echoing rejected input values or validator context.
Rate and tenancy controls API-key controls and configured member or organization limits run before JSON parsing. Audit retrieval requires the originating API key.

Implementation is the start of assurance—not evidence of production fitness.

The following staged program is proposed. It is not a claim that these activities are complete.

PHASE 1

Calculation assurance

- Independent golden vectors
- Threshold boundary tests
- Property-based invariants
- Spreadsheet/reference comparison
- Methodology-version regression

PHASE 2

Schema + provenance

- Malformed JSON fuzzing
- Source/date mismatch tests
- Coercion and float rejection
- Error redaction verification
- Cross-key receipt isolation

PHASE 3

Privacy evaluation

- Labeled detector corpora
- Precision and recall by entity
- Obfuscation variants
- Financial false positives
- Policy regression fixtures

Methodology, operations, and institutional adoption.

PHASE 4

Methodology review

- Qualified practitioner challenge
- Weights and threshold review
- Population and use boundaries
- Misinterpretation testing
- Change-approval governance

PHASE 5

Operational readiness

- Load and failure-mode tests
- Security and threat modeling
- Surrounding-log review
- Recovery and rollback
- Incident response

PHASE 6

Institution validation

- Legal and compliance review
- Representative UAT
- Reviewer workflow testing
- Controlled pilot
- Ongoing monitoring

Known boundaries should travel with the output.

Portfolio

- No covariance, volatility, drawdown, liquidity, tax, fee, or look-through modeling.
- Caller classifications and comparable positive values are assumed.
- Screening thresholds are policy indicators, not recommendations.

Risk

- Five caller-scored dimensions with fixed weights.
 - No psychometric validation or adaptive scoring.
 - Cannot be used alone for suitability or product selection.
-

Analytical and platform exclusions.

Company

- No forecasts, peer context, accounting normalization, restatement handling, or one-time adjustments.
- Multiples are omitted where inputs or positive denominators are unavailable.
- Point-in-time arithmetic is not a valuation opinion.

Platform

- Detector coverage is policy-based and should be evaluated against each institution's representative data.
- Sources are attributed but not independently verified.
- Initial agents do not browse, stream, trade, or invoke external tools.

CONCLUSION

Useful automation can be narrow, explicit, and accountable.

Grandice demonstrates a deliberately constrained approach to financial agents: typed workflow dispatch, deterministic arithmetic, caller-supplied provenance, best-effort privacy residual checks, metadata-only domain receipts, and an explicit professional-review boundary.

Its value proposition is controlled analytical consistency and integration—not autonomous judgment. Institutional deployment still requires use-case-specific validation, governance, security review, legal and compliance assessment, methodology approval, and qualified human supervision.

Response invariants and implementation references

```
{
  "request_id": "dom...",
  "privacy": {
    "mode": "client",
    "residual_check": {
      "status": "passed",
      "coverage": "configured_policy_detectors",
      "guarantee": "best_effort"
    },
    "content_retained": false
  },
  "data": {
    "methodology_version": "grandice-domain-1.0.0",
    "assumptions": [...],
    "limitations": [...],
    "warnings": [...],
    "professional_review_required": true
  }
}
```

<code>grandice_domain/common.py</code>	Shared contracts, decimal bounds, disclosures, methodology version
<code>grandice_domain/portfolio.py</code>	Portfolio arithmetic and concentration rules
<code>grandice_domain/risk.py</code>	Weighted score and band methodology
<code>grandice_domain/company.py</code>	Historical ratios, leverage, and supported multiples
<code>grandice_domain/agents.py</code>	Versioned agent registry and authority declarations
<code>gateway/domain_runtime.py</code>	Request bounds, privacy check, audit write, response envelope

IMPORTANT NOTICE

This document describes the September 2026 initial implementation. It is technical product documentation, not legal, regulatory, investment, accounting, tax, or compliance advice. Capabilities, controls, and methodology may change under future versioning. Adopting institutions must perform their own review and validation.